

A2 — TEC Heating and Cooling Analysis

Assignment code: A2

Module: PHYS39 Module 4

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1. Part 4 graph

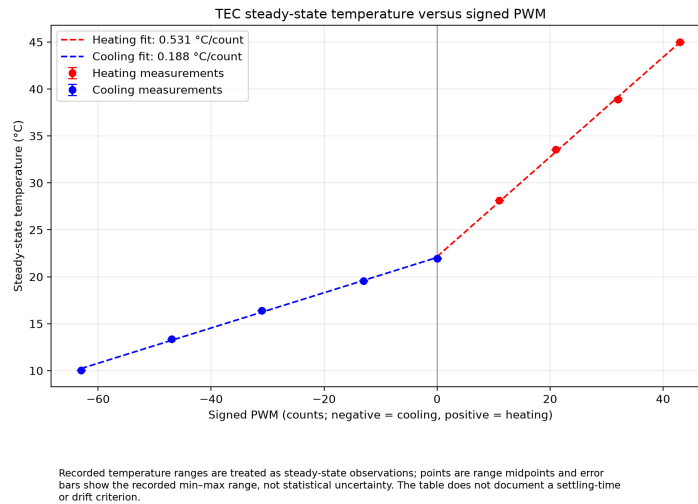


Figure 1. Heating: 0 to +43; cooling: -63 to 0 PWM counts. Points are recorded range midpoints; bars show min-max ranges. Steady-state settling/drift criteria were not documented.

2. Measured slopes and ratio

Signed-PWM slopes: $m_h = 0.531 \text{ °C/count}$; $m_c = 0.188 \text{ °C/count}$. Their ratio is $r = 0.531/0.188 = 2.8245 \approx 2.82$, using rounded graph labels. Both branches are nearly linear with slight local-slope variation.

3. PWM proof and steady-state slope derivation

During period τ , signed current is I for $D\tau$ and zero otherwise, with nonnegative duty cycle D . Direct integration gives

$$\langle i \rangle = \frac{1}{\tau} \int_0^\tau i(t) dt = \frac{ID\tau}{\tau} = DI$$

$$\langle i^2 \rangle = \frac{1}{\tau} \int_0^\tau i^2(t) dt = \frac{I^2 D\tau}{\tau} = DI^2$$

Since $\langle i \rangle^2 = D^2 I^2 \neq \langle i^2 \rangle$, Peltier and Joule terms both scale with D at fixed ON-current, consistent with the nearly linear graph. Using $\langle i \rangle^2$ incorrectly predicts quadratic Joule heating.

With thermal capacitance C , zero-PWM temperature T_0 , and passive conductance G (including TEC conduction once):

$$C \frac{dT_o}{dt} = \dot{Q}_{\text{TEC}} - G(T_o - T_0)$$

$$\frac{dT_o}{dt} = 0 \Rightarrow G(T_o - T_0) = \dot{Q}_{\text{TEC}}$$

At steady state, net heat flow vanishes. For positive ON-state Peltier power P and object-face Joule power J , u = signed PWM/255 and $\dot{Q}_{\text{TEC}} = uP + |u|J$:

$$T_h - T_0 = \frac{u(P+J)}{G} \quad (u \geq 0); \quad T_c - T_0 = \frac{u(P-J)}{G} \quad (u \leq 0)$$

$$\frac{dT_h}{du} = \frac{P+J}{G}; \quad \frac{dT_c}{du} = \frac{P-J}{G}$$

Both derivatives are positive for net cooling ($P > J$). Signed-PWM slopes each contain $1/255$, which cancels along with G :

$$r = \frac{P+J}{P-J} \Rightarrow (r-1)P = (r+1)J$$

$$\frac{J}{P} = \frac{r-1}{r+1} = 0.4771; \quad r = 2 \Rightarrow \frac{J}{P} = \frac{1}{3}$$

This assumes equal ON-current and G in both directions and constant properties.

4. Laird values, conditions, and predicted ratio

Laird CP14-127-045-L2-W4.5, part 58910-501, revision 00 (June 1, 2022), p. 3, **27 °C hot-side column:**

Quantity	Value	Meaning and condition
R	$1.50 \, \Omega$	Module electrical resistance at the listed hot-side temperature.
I_{max}	$8.6 \, \text{A}$	Current specified at ΔT_{max} .
$\dot{Q}_{\text{c,max}}$	$71.3 \, \text{W}$	Maximum cold-side heat removal at $\Delta T = 0$.
ΔT_{max}	$70.5 \, \text{K}$	Maximum face-temperature difference at $\dot{Q}_c = 0$.

At $\Delta T = 0$, passive TEC conduction vanishes. Using the assignment's symmetric half-Joule construction:

$$J_{\text{max}} = \frac{1}{2} I_{\text{max}}^2 R = \frac{1}{2} (8.6)^2 (1.50) = 55.47 \, \text{W}$$

$$P_{\text{max}} = \dot{Q}_{\text{c,max}} + J_{\text{max}} = 71.30 + 55.47 = 126.77 \, \text{W}$$

$$r_{\text{Laird}} = \frac{126.77 + 55.47}{126.77 - 55.47} = 2.5560 \approx 2.56$$

All powers are thermal; the listed maxima need not share one test condition.

5. Measured versus data-sheet ratios

Measured $r = 2.8245$ versus predicted 2.5560 : 10.5% higher. Full duty does not ensure I_{max} : supply limits, bridge drop, wiring, and TEC resistance set current. PWM versus DC, finite temperature differences, passive paths, changing properties, and fit curvature also limit agreement.

6. Passive thermal conduction

Heat flows outward when the block is hotter than room temperature and inward when colder. Symmetric

conduction opposes both excursions and cancels from r ; it cannot alone explain unequal slopes. Joule heat assists heating and opposes cooling.

7. Conclusion

The measured heating slope is $0.531\text{ }^{\circ}\text{C}$ per PWM count, while the cooling slope versus signed PWM is $0.188\text{ }^{\circ}\text{C}$ per count, giving a ratio of 2.82. Under the symmetric, fixed-current model, this implies that object-face Joule heating is about 47.7% of the Peltier heat-transfer magnitude. Peltier transport reverses with current, whereas Joule heating always adds heat, assisting heating and opposing cooling. PWM makes both contributions proportional to duty cycle, consistent with the approximately linear measured branches. Passive conduction opposes both temperature excursions and cancels from the ratio when its conductance is equal in both directions. The measured ratio is about 10.5% higher than the manufacturer-based prediction of 2.56. Different operating currents, face temperatures, and thermal conditions limit this comparison; the result is a model-based inference rather than a direct heat-flow measurement.